scenarios that could be used to categorize those problems and the best structures you could use to attack them.

Keep in mind that some topics are quite advanced and, when we delve into the details, it might require a few reads to understand everything. The book is structured in such a way to provide many levels of in-depth analysis, with advanced sections generally grouped together towards the end of each chapter, so if you'd like to get only an understanding of the topics, you are not required delve into the theory.

1.1.1 Algorithms and Data Structures: is there a difference?

Yes, they are not exactly the same thing, technically they are not equivalent. Nevertheless, we might sometimes use the two terms interchangeably and, for the sake of brevity, use the term **data structure** (*DS*) to intend "a DS and all its relevant methods".

There are many ways to point out the difference between the two terms, but I particularly like this metaphor: data structures are like *nouns*, while algorithms are like *verbs*.

I like this angle because, besides hinting at their different behavior, implicitly reveals the mutual dependency between them. For instance, in English, to build a meaningful phrase, we need both nouns and verbs, subject (or object) and action performed (or endured).